

RAMCO Institute of Technology, Rajapalayam

Department of Mechanical Engineering

Academic Year 2020 – 2021 (ODD Semester)

Innovative Practice(s) Description

Name of the Faculty : Mr. J.Jerold John Britto AP(Sr.Gr.)/Mech
Subject Code and Title : OML 751 Testing of Materials
Year and Branch : IV year Civil, EEE and ECE
Topics Covered : Unit -II: Mechanical Testing –Shear Test
Name of the Laboratory : Strength-of-Materials lab
Website Link : <http://sm-nitk.vlabs.ac.in/exp7/index.html>
Date : 12.09.2020
No.of Students participated : 60
Assignment given if any : -
(If Yes share the uploaded folder link)

Demonstration of Shear Test

OBJECTIVE

To determine experimentally, the ultimate shear strength in double shear of mild steel rod.

In direct shear test, the shearing stress is considered as uniformly distributed over the entire cross section. The shear force is applied by a suitable test rig, two different cases of shearing may arise; i.e., single shear and double shear. In single shear shearing occurs across a single surface and in double shear shearing occurs across two surfaces. Knowledge of shear failure is important while designing any structures or machine components. Shear force causes the surface to go out of the alignment with each other and thus the material fails.

The students can able to experience the shear test experimental set up in Universal Testing Machine and types of material can be identified to be tested.

Course Outcome / Programme Outcomes	PO1	PO2	PO3	PO4	PO5	PO8	PO9	PO10	PO12
CO 2 Analyze engineering components using various mechanical testing procedure.	2	2	1	2	1	2	2	2	3

The image displays two screenshots of the Virtual Labs interface for the Direct Shear Test on Mild Steel Rod.

Top Screenshot: The interface shows the 'STRENGTH-OF-MATERIALS LAB' section. The 'Direct Shear Test on Mild Steel Rod' is selected. The 'Theory' section is active, explaining that in a direct shear test, the shearing stress is considered as uniformly distributed over the entire cross section. The shear force is applied by a suitable test rig, and two different cases of shearing may arise: single shear and double shear. The interface includes navigation links: THEORY, PROCEDURE, SELF EVALUATION, SIMULATOR, ASSIGNMENT, QUIZ, VIDEO, and REFERENCE.

Bottom Screenshot: The interface shows a diagram of a universal testing machine with shear attachments and vernier calipers. The text 'Universal testing machine, Shear attachments, Vernier calipers.' is displayed. Below the diagram, it says '© 2016 - 2020 SOLVE - The Virtual Lab @ NITK Surathkal, Department of Water Resources & Ocean Engineering'. A 'DIRECT SHEAR TEST ON MILD STEEL ROD' title bar is visible at the bottom of the simulation area.

The screenshot shows a virtual lab environment for a Universal Testing Machine (UTM). The main window displays a diagram of the machine with a control panel on the left featuring a large circular pressure gauge and several smaller gauges. A specimen is shown being tested in the center. The text "STEP 3" is prominently displayed, indicating the machine is switched ON and the load is gradually increased until the specimen is fractured. Below the diagram, the failure load is noted as $P = 1350\text{Kg}$. The interface also shows "TRIAL : 1" and a copyright notice for NITK Surathkal. The bottom of the screen shows a Windows taskbar with various application icons and a system clock.